

RRDetection: Respiration Rate Estimation Using Earbuds During Physical Activities

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Abstract—In modern times, earbuds have become both popular and essential accessories for people to use with a wide range of devices in their everyday lives. Moreover, the respiration rate is a crucial vital sign that is sensitive to various pathological conditions. Many earbuds now come equipped with multiple sensing capabilities, including inertial and acoustic sensors. These sensors can be used by researchers to passively monitor users' vital signs, such as respiration rates. While current earbud-based breath rate estimation algorithms mostly focus on resting conditions, recent studies have demonstrated that respiration rates during physical activities can predict cardio-respiratory fitness for healthy individuals and pulmonary conditions for respiratory patients. To address this gap, we propose a novel algorithm called *RRDetection* that leverages the motion sensors in ordinary earbuds to detect respiration rates during light to moderate physical activities.

I. INTRODUCTION

Despite the availability of multiple wearable sensing techniques for measuring respiration rate, such as earbuds [1] and smartphones [2], monitoring respiration rate during physical exercise is uncommon due to the overlap in frequency ranges between body motion and breathing rate. However, the significance of monitoring respiration rate is increasing [3], [4]. For example, cardio-respiratory fitness (CRF) refers to the ability of the circulatory and respiratory systems to supply oxygen to skeletal muscles during sustained physical activity, and it is a crucial indicator of physical and mental health in young people [5], [6]. In the United States, only 40% of youths are believed to have healthy CRF [7]. Therefore, detecting breathing parameters during sustained physical activity to monitor CRF levels is beneficial for evaluating individuals' health and fitness [5].

Respiration rate has traditionally been measured using respiratory belts [8] or impedance pneumography. Nevertheless, these methods are expensive and inconvenient to use outside clinical settings [9]. Recently, some studies have demonstrated the feasibility of measuring respiration rate outside of clinical settings using consumer devices [1], [10], [11]. For instance, Ahmed [1] used a wireless earphone with mounted IMU sensors to monitor breath rate, while Nam *et al.* [12] utilizes a microphone placed under the nose to estimate the breath rate. However, these respiratory monitoring devices are generally employed when a person is at

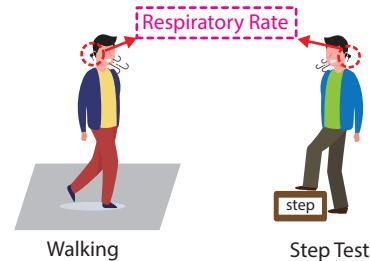


Fig. 1. RRDetection system measures breath rate using earbud's in-built inertial motion sensors while the users wear them during physical activities.

rest, and there are few methods for estimating breathing rate during physical activities. Recently, Kumar *et al.* [10] utilizes wearable microphones to estimate breathing rate. The authors use a multi-task LSTM network with convolutional layers to determine respiration rate in noisy environments before, during, and after a workout session. Nonetheless, acoustic sensing can be resource-intensive and privacy sensitive in the real-world settings.

In this work, we propose *RRDetection* which utilizes the built-in motion sensors (IMU) from commercial earbuds to monitor respiration rate during physical activities. Our algorithm dampens the physical activity signal (e.g., steps) and detects the underlying breathing signal. Compared to acoustic sensing, motion sensor-based approaches consume lower resources and have the potential to continuously monitor the user's breathing rate during the long duration of physical activities. We demonstrate that *RRDetection* can track overall breathing cycles during light to moderate physical exercises (Figure 1) using motion sensor data from commercial earbuds. We conducted two data collection studies to evaluate the performance of our approach with two types of physical activities commonly used for cardio-respiratory fitness tests. We compare the performance of the proposed method with the traditional peak detection-based algorithm for breathing rate estimation. Our algorithm outperforms the traditional peak detection-based algorithm in terms of mean absolute error (MAE), achieving a lower MAE of 3.87 breaths per minute (compared to 4.83 breaths per minute for peak-based MAE) during the 3-minute Step Test and a lower MAE of 3.75 breaths per minute (compared to 5.21 breaths per minute for peak-based MAE) during the 6-minute Walk Test.

¹ The subject matter of this paper was developed while the author was doing internship at Samsung Research America.

II. RELATED WORKS

In general, respiratory activity sensing can be classified into three broad categories. The first category is IMU-based breath activity detection in a static environment. Cesareo *et al.* [13] utilizes IMU sensors to monitor the breath rate. The second category is acoustic-based breath rate calculation. Nam *et al.* [12] estimated the respiration rate by using a microphone placed under the nose. Islam *et al.* [2] placed smartphone around the chest to capture breathing sound from the nose and mouth. The third category is chest displacement sensing for breath rate estimation. Song *et al.* [14] and Hernandez *et al.* [15] measured distance changes during a breathing activity using a smartphone.

Moreover, some recent wearable techniques are employed to measure human respiration rates or breathing events, such as apnea [10], [16], [17]. However, most previous studies have focused on breath rate estimation during rest or with power-hungry audio. In contrast, we propose a novel method for sensing breathing rate during physical activities using an earbud equipped with inertial sensors.

III. SYSTEM DESIGN

A. Challenges and Solutions

It is well acknowledged that a person's breath signals collected from earbuds are correlated with small variations in her/his breath rate intervals, the dominant part of the IMU signals from earbuds is correlated to exercises as shown in Fig. 2(a). Therefore, in detecting breath rate, the main challenge is that the morphology of breath signals is unclear and inconsistent during exercises. To address this challenge, our research goal is to dampen the activity signals and improve the quality of the breath signals as shown in Fig. 2(a) and Fig. 2(b). Therefore, we proposed the *RRDetection* solution which consists of mainly two components, one is for extracting corresponding breath signals from raw step signals (i.e., data selection and smoothing, data acceleration, two-stage filter), and the other component is to calculate breath rate by segmenting the extracted signals into breath cycle and then counting the number of breaths in a minute. As shown in Fig 2(c), the black dashed line stands for the segments which refer to the breathing cycles.

B. *RRDetection* System

As shown in Fig. 3, *RRDetection* consists of five components to estimate respiration rate.

1) *Data Selection and Smoothing*: The respiratory rate of an individual increases as they engage in physical exercises. To capture this, we selected the data from the last minute of the ongoing step test, as it has a higher respiration intensity. To make our algorithm independent of earbud orientation inside the ear, we calculated the respiration rate using the accelerometer magnitude $\sqrt{x^2 + y^2 + z^2}$, similar to our previous work [1].

2) *Step Signal Dampening*: When someone is performing physical activity such as walking or a step test, the earbud motion sensor captures both the step signal and the underlying breathing signal. However, the step signal is more dominant than that of breathing. Therefore, our system needs to dampen the step signals and improve the signal-to-noise ratio (SNR) for breathing signals. To achieve this, we utilized the following numerical method, which is widely used in different applications [18], [19], to dampen the raw step signals:

$$f'' = \frac{4f_0 + (f_1 + f_{-1}) - 2(f_2 + f_{-2}) + (f_3 + f_{-3})}{16h^2} \quad (1)$$

Here, f_i refers to the value of the time series i samples away, and h is the time interval between consecutive samples. After obtaining the dampened signals from the step signals, we performed a two-stage filtering process to denoise the signal. Firstly, we applied a 4th-order difference filter. Next, we used a z-score-based outlier filter to reduce isolated noise spikes. These two-stage filters allowed us to obtain a stable signal for extracting breath cycles.

3) *Segmentation*: The breath cycles have similar patterns, and we need to calculate the number of breath cycles in a minute to determine respiration rate. To obtain individual breath cycles, we adopted and optimized a segmentation algorithm that is commonly used for extracting heartbeats from wireless signals [18]. In the following section, we provide a detailed explanation of our optimized algorithm 1. The basic idea is to find a template that is most similar to each breath cycle.

a) *Inputs and Outputs*: We take the extracted breath signals, $x = (x_1, x_2, \dots, x_n)$, as input, and the output is a set of segments, $S = (s_1, s_2, \dots, s_k)$. We also define μ as the template that we aim to find, which is most similar to each segment $(s_1, s_2, \dots, s_k)$. This can be expressed as $s_1 \approx \mu, s_2 \approx \mu, \dots, s_k \approx \mu$. Since the possible breath cycle ranges from 12 BPM to 40 BPM, we constrain the length of each segment by setting $min_{cycle} = 12$ and $max_{cycle} = 40$.

b) *Update Segmentation and Template*: Our algorithm first updates the segmentation and then updates the template during each iteration. In current iteration, segmentation S^{i+1} is updated based on the previous template μ^i as $S^{i+1} = argmin \sum Dist(s_k, \mu^i)$, where $Dist()$ stands for the distance between each segment (s_k) and the template (μ^i) . To get the most accurate distance between two signals, we utilized the cross-correlation to calculate the distance between each segment and template from the time domain (i.e., $xcorr(s_k, \mu)$), frequency (i.e., $xcorr(fft(s_k), FFT(\mu))$), and signal variance (i.e., $xcorr(s_k^2, \mu^2)$) with specific weight 0.5, 0.2 and 0.3, respectively. Then, we update the template $\mu^{i+1} = argmin \sum Dist(s_k, \omega(\mu_i))$.

In our algorithm, it is necessary to search the samples from extracted breath signals to calculate the distance between each segment and the template. And, we trade off the computation cost and accuracy by setting a configurable step length (i.e., *step_len*), which means searching the signals with a batch size of points. Besides, to compare

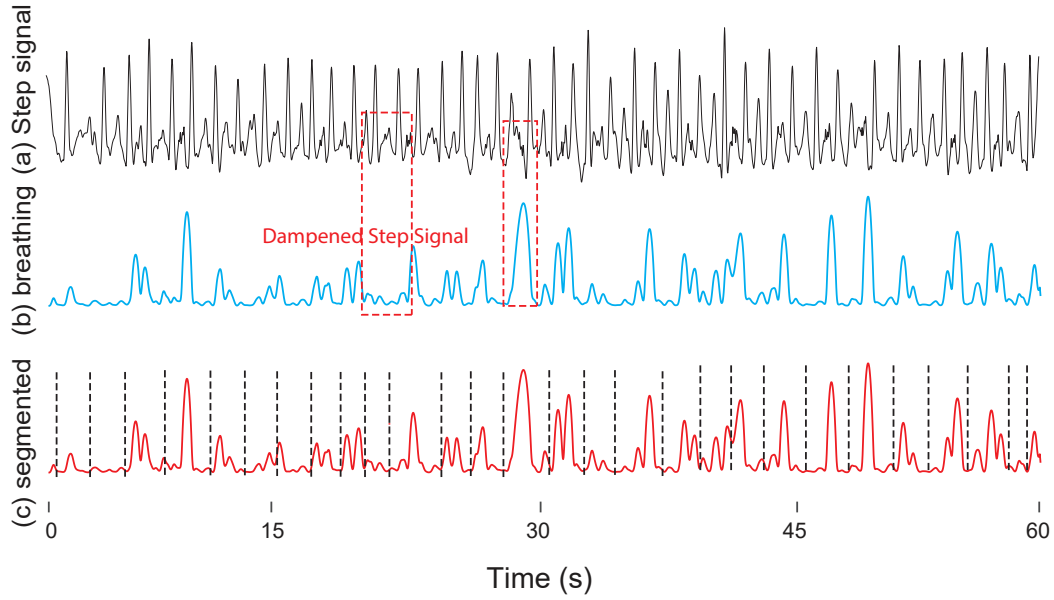


Fig. 2. Example of processing flow from raw accelerometer signal to extracted breathing signals for estimating breath rate based on our optimized segmentation algorithm. (a) Raw IMU signals during exercises. (b) The extracted breathing signals. (c) Breathing cycle segmentation using the proposed algorithm on the extracted breathing signal.

signals, we deployed an algorithm to measure the similarity. The method is to calculate the total distance between two signals from three aspects (i.e., time domain analysis, time-frequency analysis, and signal variance analysis). The details of the segmentation part based on the similarity metric are explained in Algorithm 1.

4) *Respiration Rate Calculation*: Since we chose one-minute data as our window size, respiration rate is simply the number of segments in the window.

IV. ALGORITHM PERFORMANCE EVALUATION

A. Participants and Data Collection

To validate our proposed method, we conducted two data collection studies with two different types of physical activities. The first study was conducted with 20 subjects, and the participants performed a step test in lab and at home for three minutes (Step-Test Study). The second study was conducted with 30 subjects, where the subjects performed walking in regular pace for six minutes (Walking Study). During the data collection, the participants wore a pair of earbuds and a Zephyr Bioharness Biomodule3 respiratory belt as a reference device. The respiratory belt provides the reference breathing waveform. We annotated the breathing waveform and convert them into ground truth respiration rate by counting the breathing cycles in each minute. The 6-axis motion sensor data was collected at a 48Hz sampling rate. Both studies were approved by Institutional Review Board.

B. Evaluation

In this section, we evaluate the performance of the RRDetection system using different datasets, including 3-Minute Step Tests conducted in the lab and at home, as well as 6-Minute Walk Test data. Since our algorithm is designed to estimate breathing rate in the 12-40 breaths per minute

(BPM) range, we only considered ground truth values within this range. To validate the effectiveness of our RRDetection algorithm in estimating respiration rate, we compared it with the traditional peak-based algorithm (i.e., using 'find-peaks' function with a parameter 'MinPeakDistance' of $1.75 \times \text{sample rate}$ and 'MinPeakHeight' of 0.1). To evaluate the agreement between the RRDetection algorithm and the ground truth, we utilize Bland-Altman plots.

C. Breathing Rate Detection During Exercises

1) *3-minute Step Test Dataset*: The cardiorespiratory fitness level of an individual can be measured using a 3-minute step test, which involves stepping on and off a 12-inch bench for 3 minutes while maintaining a consistent pace [20]. The test requires setting the metronome to 96 beats per minute, and subjects start marching up and down the bench when the clock or stopwatch is started. We conducted the 3-minute step tests in both lab and home settings, and analyzed the results independently.

a) *Step Test in Lab*: As shown in Table I, the MAE and MAPE of segmentation-based respiration rate detection are 3.3 breaths per minute and 13.6%, respectively, while the MAE and MAPE of peak-based breath rate detection are 3.84 and 14.7% respectively. The average baseline reference breath rate is 24.4 BPM. To further analyze the breath rate, we divided the data into two groups, the first group is called the Near REF group (i.e., $\frac{\text{current_subject_ref} - \text{average_ref}}{\text{average_ref}} \leq 20\%$). All other subjects belong to the other group which is called Far REF group. We can observe that the MAE of our proposed segmentation method in the Far REF group is much better than the peak-based algorithm. This result shows our algorithm is performing better to estimate the breath rate when the breath rate is deviated from the baseline breathing rate.

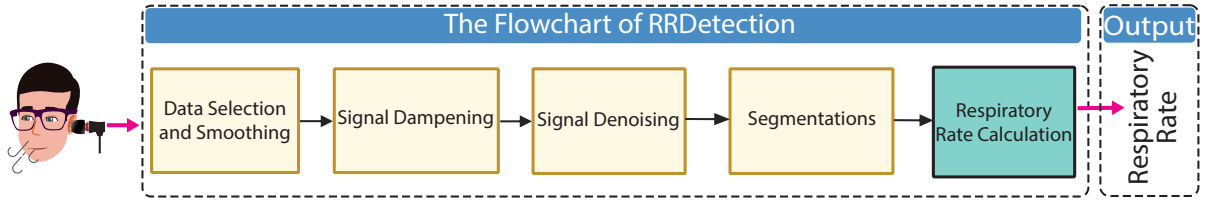


Fig. 3. RRDetection system overview for data processing and breathing rate estimation during activities.

Algorithm 1 Breathing Segmentation Algorithm

Require: Do segmentation after doing signal optimization

Input: sequence x for breath signals.

Output: Segments S , template μ .

Initialize μ as a zero vector.

$l \leftarrow 0$ {Initialization iteration}

Repeat { * Update segmentations and template together
* }

$S^{l+1} \leftarrow \text{UpdateSegmentation}(x, \mu^l)$

$\mu^{l+1} \leftarrow \text{UpdateTemplate}(x, S^{l+1})$

Return S^l and μ^l

Procedure $\text{UpdateSegmentation}(x, \mu^l)$

$S_0 \leftarrow \emptyset$

$D_0 \leftarrow 0$

for $t \leftarrow 1$ to n **do**

$\tau \leftarrow \text{argmin}_{\tau \in \tau_t, B} D_\tau + \text{Dist}(x_{\tau+1:t} - \omega(\mu, t - \tau))$

$D_t \leftarrow D_{\tau^*} + \text{Dist}(x_{\tau^*+1:t} - \omega(\mu, t - \tau))$

$S_t \leftarrow S_{\tau^*} \cup x_{\tau^*+1:t}$

end for

End Procedure

Procedure $\text{UpdateTemplate}(x, S^l)$

$\mu \leftarrow \frac{1}{n} \sum_{s_i \in S} |S_i| \omega(s_i, m)$

End Procedure

TABLE I

ALGORITHM PERFORMANCE ON 3-MIN STEPS TEST LAB DATA.

Algorithm	MAE	MAPE	MAE Near REF	MAE Far REF
Peak_based	3.84	14.7%	2.61	7.55
Segmentation	3.30	13.6%	2.96	4.35

b) Step Test at Home: We also conducted 3-minute step tests at home. As shown in Table II, the MAE and MAPE of segmentation-based respiration rate detection are 3.87 breaths per minute and 20.2%, respectively. In contrast, the MAE and MAPE of peak-based breath rate detection are 4.83 and 25.7%. The average reference breath rate is 21.9 BPM. Still, based on the MAE of Near REF group and the MAE of Far REF group, we can observe that our proposed segmentation-based algorithm is better than peak-based algorithm.

2) *6-minute Walk Test Dataset at Lab:* Walking is one of the most common physical activities. Thus, we experimented with a 6-minute walking test to evaluate the performance of

TABLE II

ALGORITHM PERFORMANCE ON 3-MIN STEPS TEST HOME DATA.

Algorithm	MAE	MAPE	MAE Near REF	MAE Far REF
Peak_based	4.83	25.7%	3.15	7.44
Segmentation	3.87	20.2%	2.31	6.7

TABLE III

ALGORITHM PERFORMANCE ON 6-MINUTE WALK TEST DATA

Algorithm	MAE	MAPE	MAE Near REF	MAE Far REF
Peak_based	5.21	19.1%	2.86	12.28
Segmentation	3.75	15.9%	2.96	4.55

our proposed algorithm. As shown in Table III, the MAE and MAPE of peak-based breath rate detection are 5.21 BPM and 19.1%, respectively, while the MAE and MAPE of segmentation-based breath rate detection are 3.75 BPM and 15.9%. The average reference breath rate is 23.3 BPM. Based on the MAE of Near or Far REF group, we can observe that the MAE of our proposed segmentation method (i.e., 4.55 BPM) in the Far REF group is much better than the peak-based algorithm (i.e., 12.28 BPM).

D. Consistency Evaluation

We used a Bland-Altman plot to measure the agreement between two measurement methods (i.e., segmentation-based vs. ground truth, peak-based vs. ground truth), which is a useful tool to visualize fixed bias, proportional bias, and limits of agreement [21].

As shown in Fig. 4, for the 3-minute step tests in the lab dataset, we observed a fixed bias of -0.08 BPM using the segmentation-based algorithm, with a limit of agreement between -8.9 and 8.7 . In contrast, we observed a fixed bias of 2.9 BPM using the peak-based algorithm, with a limit of agreement between -5.8 and 11 . Furthermore, Fig. 5 shows the agreement between the reference and predicted breath rate for the 6-minute walk dataset. We observed a fixed bias of -0.13 breaths per minute and limits of agreement between -11 and 11 using the segmentation-based algorithm, while we observed a fixed bias of 2.0 breaths per minute and limits of agreement between -14 and 18 using the peak-based algorithm. Based on these results, we can conclude that the segmentation-based estimation method performs more consistently across different breathing rates and is more suitable as a standalone algorithm in the earbuds.

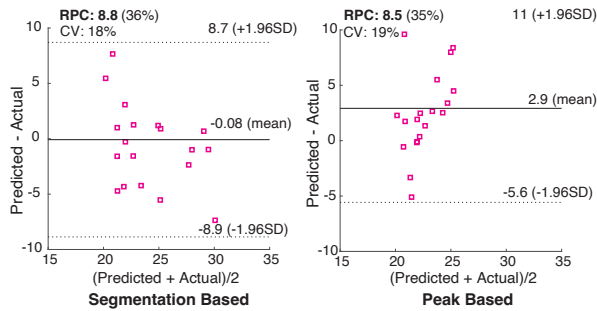


Fig. 4. Bland-Altman Plot showing the agreement between the ground truth and predicted breath rates based on 3-minute step tests in lab.

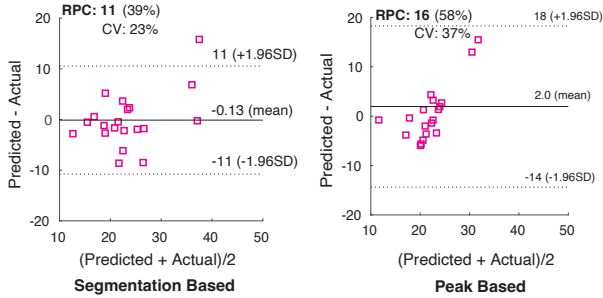


Fig. 5. Bland-Altman Plot showing the agreement between the ground truth and predicted breath rates on 6-minute walk tests.

V. CONCLUSIONS AND DISCUSSIONS

This work RRDetection presents a respiration rate estimation method during activities (e.g., 3-minute step test and 6-minute walk test) which can help predict cardio-respiratory fitness. Based on the performance evaluation on two different datasets, our algorithm shows effectiveness in estimating respiration rate during exercises by using earbud's accelerometer data. Since the earbud is becoming increasingly popular consumer wearables, the algorithm presented in this paper has a great potential to estimate cardio-respiratory fitness of the user and monitor the fitness trend over time without additional burden from the users.

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